

The Higgs–vacuum route to ξ

Origin, formulas and status of a consistency check

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The Higgs–vacuum route to ξ

Abstract

The first route to the relations of FFGFT ran through the vacuum and the Higgs field. In spring 2025 it produced a formula linking the Higgs quantities to the electric constant ε_0 : $\lambda_h^2 v^2 e^2 / (64\pi^4 \varepsilon_0 \hbar c m_h^2)$. This document traces the formula through the version history from February 2025 to today, recomputes every stage and states the current status. Written out, the vacuum formula is exactly $\xi_{\text{EFT}} \cdot \alpha$ with $\xi_{\text{EFT}} = \lambda_h^2 v^2 / (16\pi^3 m_h^2)$; with the FFGFT convention $\alpha = 1$ it becomes today's Higgs route. With the self-coupling $\lambda_h = m_h^2 / (2v^2)$ one obtains $\xi_{\text{EFT}} = m_h^2 / (64\pi^3 v^2) = 1.30 \times 10^{-4}$, about 2.3 % below $\xi = 4/30000$. The Higgs geometry is therefore a consistency check of ξ to within a few per cent **[K]**, not an exact derivation. Not sustainable are the early equating of the expression with 1, a short-lived variant with μ_0 / ε_0 , and the later shortcut to $1/(16\pi^3)$ **[X]**. The factor $64\pi^4$ of the early HTML versions comes from the 4π in $4\pi\varepsilon_0$.

Status markers

- [B]** Algebraically proven.
- [K]** Numerically confirmed (check script, comparison with measured values).
- [S]** Assumption, open or speculative.
- [X]** Refuted or not sustainable.
- [Q]** Convention.

Reference values: CODATA 2022, PDG 2024 ($m_h = 125.20$ GeV, $v = 246.22$ GeV). The sources of the early formulas are commits of the repository; the files now exist only in the version history. Check script: `pruef_385_higgs_vakuum.py`.

1 Starting idea: vacuum, Higgs field and the speed of light

In February 2025 the starting point was a qualitative thesis: the vacuum is not empty, and the Higgs field together with the background radiation determines the maximal speed. The link ran through the vacuum constants,

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}, \quad Z_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 376.73 \, \Omega, \quad (1)$$

still without a formula for a parameter (first upload of 27 March 2025, document on the Schwarzschild metric). In parallel, notes appeared expressing μ_0 and ϵ_0 through α , e and c , $\alpha = e^2/(4\pi\epsilon_0\hbar c)$. These two lines, Higgs and vacuum on one side, vacuum constants and α on the other, are the origin of the route traced here.

2 The chain of formulas in 2025

Date	Formula	Finding
31 Mar 2025	$\beta = \frac{\lambda_h^2 v^2}{16\pi^3 c^3} \cdot \frac{\hbar}{m_h^2} \cdot \frac{1}{r_0}$	factor $16\pi^3$ appears; not dimensionless in SI
5 Apr 2025	$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.33 \times 10^{-4}$	correctly evaluated 1.30×10^{-4}
6 Apr 2025	$\frac{\lambda_h^2 v^2 e^2}{64\pi^4 \epsilon_0 \hbar c m_h^2} = 1$	$= \xi_{\text{EFT}} \alpha$, not 1
18 Apr 2025	$\beta_T = \frac{\lambda_h^2 v^2}{4\pi^2 \lambda_0^2 \alpha_0}$, dimension $(\mu_0/\epsilon_0)^k$	dimensionally inconsistent
25 May 2025	ϵ_0 eliminated via $e^2 = 4\pi\epsilon_0$; $64\pi^4 \rightarrow 16\pi^3$	return to the Higgs expression
from 1 Jun 2025	$\xi = \frac{\lambda_h^2 v^2}{64\pi^4 m_h^2} = 1.04 \times 10^{-5}$	4π from $4\pi\epsilon_0$ left over
22 Aug 2026	$\lambda_h = m_h/v \Rightarrow \xi = 1/(16\pi^3)$, iteration to 4/30000	shortcut, not sustainable

The Higgs expression

On 31 March 2025 an expression with the Higgs quantities and the factor $16\pi^3$ appears for the first time, then as a parameter β with a radius r_0 . On 5 April 2025, with $r_0 = \xi \ell_P$, natural units and $\beta_T = 1$, it becomes

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2}. \quad (2)$$

The document gives 1.33×10^{-4} ; computed from the inputs it is 1.30×10^{-4} . The same document already contains the rewriting with $m_h^2 = 2\lambda_h v^2$: $\xi = \lambda_h/(32\pi^3)$. The expression is dimensionless and correctly formed **[B]**.

The vacuum formula

On 6 April 2025 the document connects, via natural units with $\alpha = 1$, the Higgs expression with the fine-structure constant in its SI form:

$$\beta_T \alpha_{\text{EM}} \approx \frac{\xi \lambda_h^2 v^2}{16\pi^3 m_h^2} \cdot \frac{1}{\xi} \cdot \frac{e^2}{4\pi\epsilon_0 \hbar c} = \frac{\lambda_h^2 v^2 e^2}{64\pi^4 \epsilon_0 \hbar c m_h^2} = 1. \quad (3)$$

The same document contains $c = 1/\sqrt{\mu_0 \epsilon_0}$, $Z_0 = \sqrt{\mu_0/\epsilon_0}$ and $h = 2\pi\hbar\mu_0 c^2$. This is the formula in which the Higgs sector and the vacuum constants appear together for the first time.

The variant with μ_0/ϵ_0

On 18 April 2025 an English working draft attempts an "electromagnetic derivation" of β_T from μ_0 and ϵ_0 : $\beta_T^{\text{nat}} = \lambda_h^2 v^2 / (4\pi^2 \lambda_0^2 \alpha_0)$ with the Higgs Compton length $\lambda_0 = \hbar/(m_h c)$ and a dimension factor $(\mu_0/\epsilon_0)^k$, $k = 0$. The file was deleted two days later and never adopted.

Return and offshoot

On 25 May 2025 ϵ_0 is eliminated via $e^2 = 4\pi\epsilon_0$ ($\alpha = 1$, $\hbar = c = 1$); $64\pi^4$ turns back into $16\pi^3$, and the expression is again the Higgs expression of 5 April. From 1 June 2025, however, the HTML versions and Doc. 049 contain $\xi = \lambda_h^2 v^2 / (64\pi^4 m_h^2) = 1.04 \times 10^{-5}$: here ϵ_0 was dropped, but the 4π from $4\pi\epsilon_0$ was not cancelled.

The shortcut of 2026

When Doc. 320 was created on 22 August 2026, $\lambda_h = m_h/v$ was inserted. Everything then reduces to $1/(16\pi^3)$, and an iteration is supposed to lead from there to the fixed point $4/30000$; A130 and Doc. 133 are cited as sources. This substitution confuses the Higgs self-coupling with a Yukawa coupling $y = m/v$; the same identification $y = \lambda_h$ already appears in Doc. 097 in 2025. A130 contains the Higgs route in no version, Doc. 133 contains no iteration.

3 What the vacuum formula means

Written out, the right-hand side of (3) is a product of two dimensionless factors:

$$\frac{\lambda_h^2 v^2 e^2}{64\pi^4 \epsilon_0 \hbar c m_h^2} = \underbrace{\frac{\lambda_h^2 v^2}{16\pi^3 m_h^2}}_{\xi_{\text{EFT}}} \cdot \underbrace{\frac{e^2}{4\pi\epsilon_0 \hbar c}}_{\alpha} = \xi_{\text{EFT}} \cdot \alpha \quad \text{[B]}. \quad (4)$$

The formula is dimensionally correct. The vacuum constant ε_0 enters only through α , and the factor $64\pi^4 = 16\pi^3 \cdot 4\pi$ arises because the 4π from $4\pi\varepsilon_0$ moves into the denominator. In SI the expression gives $\xi_{\text{EFT}} \alpha = 9.5 \times 10^{-7}$ **[K]**. With the FFGFT convention $\alpha = 1$, i.e. $e^2 = 4\pi\varepsilon_0\hbar c$, it is exactly ξ_{EFT} **[B]**. The Higgs–vacuum route and today’s Higgs route are therefore the same calculation: the vacuum constants disappear in natural units, and what remains is the Higgs expression that checks ξ . Doc. 310 reads it the same way: in natural units μ_0 and ε_0 disappear, and only ξ remains.

Not sustainable is the equating with 1 in (3) **[X]**. In the first step $\xi \cdot (\dots) \cdot (1/\xi)$ is cancelled, while β_T itself contains the factor $1/\xi$. Read correctly, $\beta_T = \xi_{\text{EFT}}/\xi = 0.977$, close to but not exactly 1, and $\beta_T\alpha = 0.0071$.

The variant with μ_0/ε_0 of 18 April is dimensionally inconsistent **[X]**: λ_0^{-2} has the dimension energy², so v^2/λ_0^2 has energy⁴; with $\lambda_0 = 1/m_h$ the expression gives about $4 \times 10^5 \text{ GeV}^4$, neither 1 nor ξ . The factor $(\mu_0/\varepsilon_0)^k = Z_0^{2k}$ appears only in the dimension line and vanishes for $k = 0$.

4 Current status

With the Standard Model self-coupling $\lambda_h = m_h^2/(2v^2) = 0.129$, the Higgs expression becomes

$$\xi_{\text{EFT}} = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} = \frac{m_h^2}{64\pi^3 v^2} = \frac{\lambda_h}{32\pi^3} = 1.30 \times 10^{-4} \quad \textbf{[K]}, \quad (5)$$

about 2.3 % below $\xi = 4/30000 = 1.333 \times 10^{-4}$. It depends only on the ratio m_h/v . Depending on the inputs the gap lies between 2.2 and 2.4 %; the frequently quoted 2.6 % arises only with λ_h rounded to 0.129:

Inputs	ξ_{EFT}	Gap to 4/30000
$m_h = 125.20$, $v = 246.22 \text{ GeV}$ (PDG 2024)	1.303×10^{-4}	−2.3 %
$m_h = 125.25$, $v = 246.22 \text{ GeV}$	1.304×10^{-4}	−2.2 %
$m_h = 125.1$, $v = 246.2 \text{ GeV}$	1.301×10^{-4}	−2.4 %
$\lambda_h = 0.129$ rounded, $m_h = 125.1$	1.299×10^{-4}	−2.6 %

The Higgs geometry is therefore a consistency check of ξ to within a few per cent **[K]** (Doc. 354, A190, 310). The deviation is interpreted structurally **[S]** (R59): the SI values of m_h and v are scheme dependent, and fractal corrections cannot be cleanly separated from the loop structure of the Standard Model. The Higgs route is not an exact derivation of 4/30000. It is, however, an independent check: it uses only the Higgs mass and the electroweak scale, no lepton mass and no Galois counting.

5 What holds and what does not

Statement	Status	Reason
Vacuum formula = $\xi_{\text{EFT}} \cdot \alpha$	[B]	written out; $\alpha = e^2/(4\pi\varepsilon_0\hbar c)$
with $\alpha = 1$ equal to the Higgs expression	[B]	$e^2 = 4\pi\varepsilon_0\hbar c$
$\xi_{\text{EFT}} = m_h^2/(64\pi^3 v^2) = 1.30 \times 10^{-4}$, about 2.3 % below ξ	[K]	consistency check
Deviation structural (scheme, fractal correction)	[S]	R59
Vacuum formula = 1	[X]	cancellation of $1/\xi$ inconsistent
Variant with μ_0/ε_0	[X]	dimension energy ⁴
$\xi = \lambda_h^2 v^2/(64\pi^3 m_h^2) = 1.04 \times 10^{-5}$	[X]	4π from $4\pi\varepsilon_0$ not cancelled
$\lambda_h = m_h/v$, $\xi = 1/(16\pi^3)$, iteration to 4/30000	[X]	15 times ξ_i ; iteration runs to 0
Higgs route as exact derivation of 4/30000	[X]	gap 2–3 %

6 Aligned places in the corpus

With this document the statements on the Higgs route in the corpus are made consistent; each place carries a dated note. The wrong numbers (1.318×10^{-4} , 1.33×10^{-4} , " $= 4/3 \cdot 10^{-4}$ ", "exact", "uniquely determined") in Docs. 006, 046, 049, 054, 061, 067, 068, 073, 175, 189 and A142 are traced back to $\approx 1.30 \times 10^{-4}$ and a consistency check; Doc. 054 listed a "CODATA value" for ξ , which does not exist. In Doc. 320 the shortcut is noted and the sources are corrected, in Doc. 354 the rounding dependence and the "fixed point" that had been listed as proven (now **[S]**), in Doc. 186 the origin of the $64\pi^4$, in Doc. 310 the reference of the "additional sphere" to the 4π from $4\pi\epsilon_0$. The check scripts T0-test now compute λ_h from m_h and v .

7 Conclusion

The Higgs–vacuum route that stood at the beginning of FFGFT holds. Its vacuum formula of April 2025 is, written out correctly, the product of the Higgs expression and the fine-structure constant; with $\alpha = 1$ it is today's Higgs route, and this route meets ξ to about 2.3 %. The errors of the line of development lie in side steps: the equating with 1, a dimensionally wrong variant, an uncanceled 4π and a late shortcut. As a derivation of the exact number $4/30000$ the route is not sufficient; as an independent check of ξ from the Higgs sector it stands.

Check script: 2/python/Dok385_Skripte/pruef_385_higgs_vakuum.py — 17/17.